AIG nodes

AIGs use only AND + optional edge complement , not arbitrary gates yet

The idea

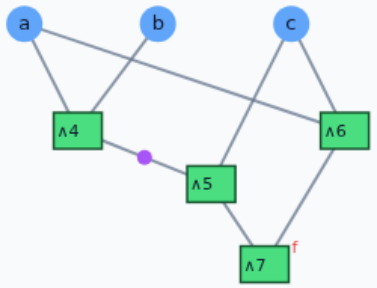
- Each AND has two fanins (id, complemented)
- Evaluate bottom-up with bitwise AND

Kinds

AIG NODES

Kinds

Step 1 / 5 · kinds · module01-03-aig-nodes



Explanation

PI circles vs AND rectangles.

- Two shapes

```
PI/AND
```

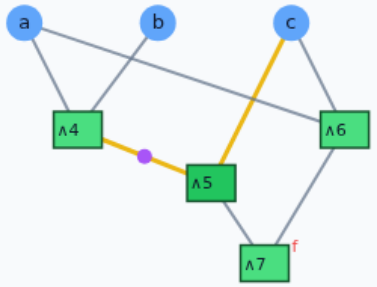
Kinds

Fanin

AIG NODES

Fanin

Step 2 / 5 · fanin · module01-03-aig-nodes



Explanation

Node 5 takes ~4 and c.

- Complement
- c

```
id 5
```

Fanin

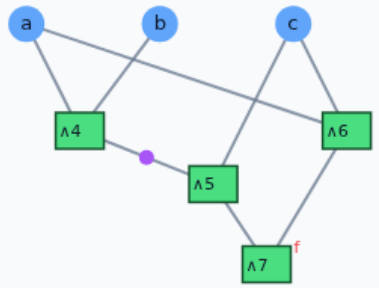
Eval

AIG NODES

Eval

Step 3 / 5 · eval · module01-03-aig-nodes

f=1



Explanation

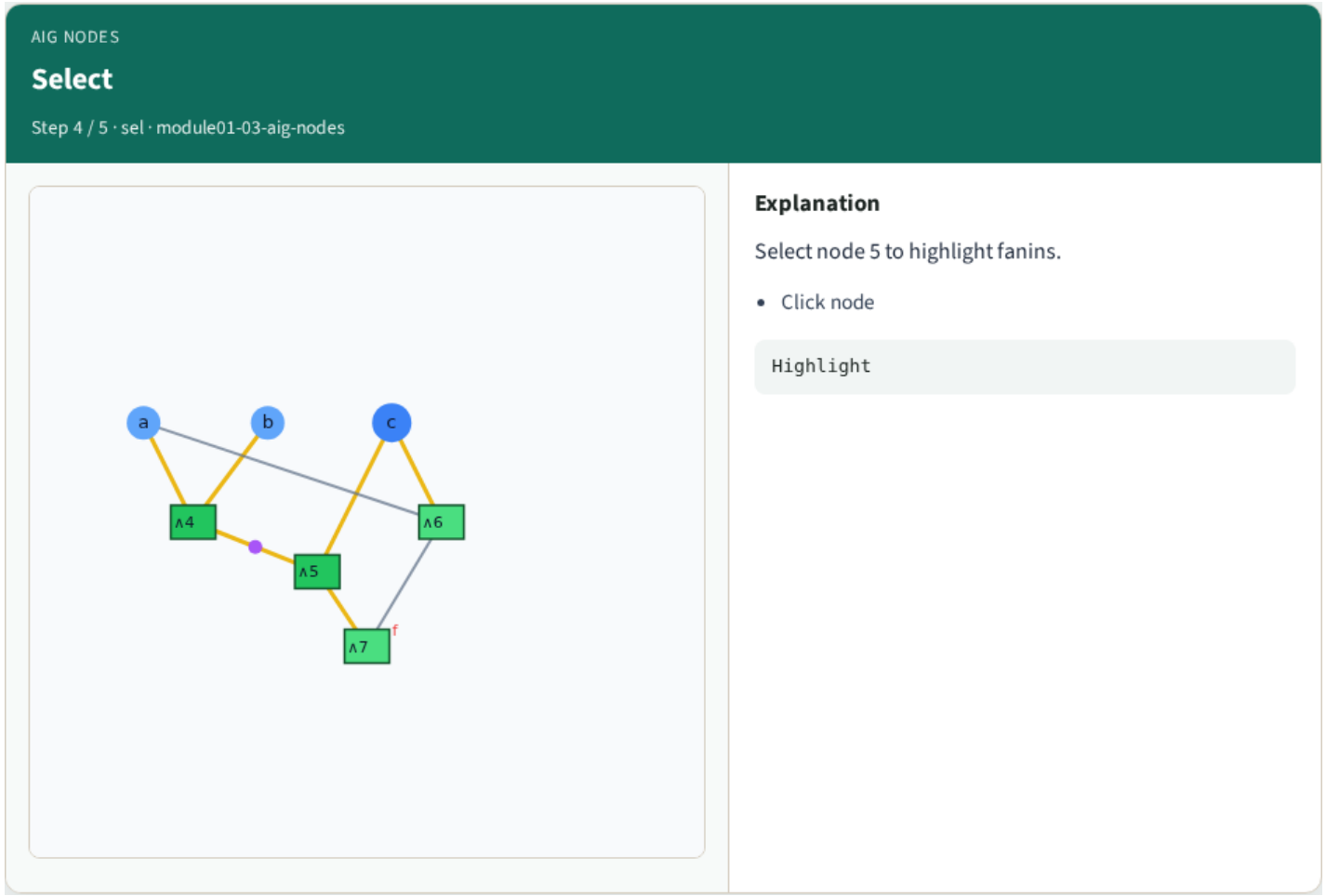
With a=1,b=0,c=1 the PO f equals 1.

- Simulate

```
f=1
```

Eval

Select



Select

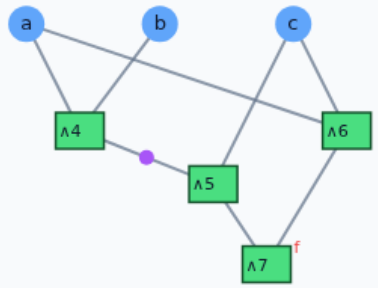
Count

AIG NODES

Count

Step 5 / 5 · count · module01-03-aig-nodes

{"pi":3,"and":4,"po":1,"total":7}



Explanation

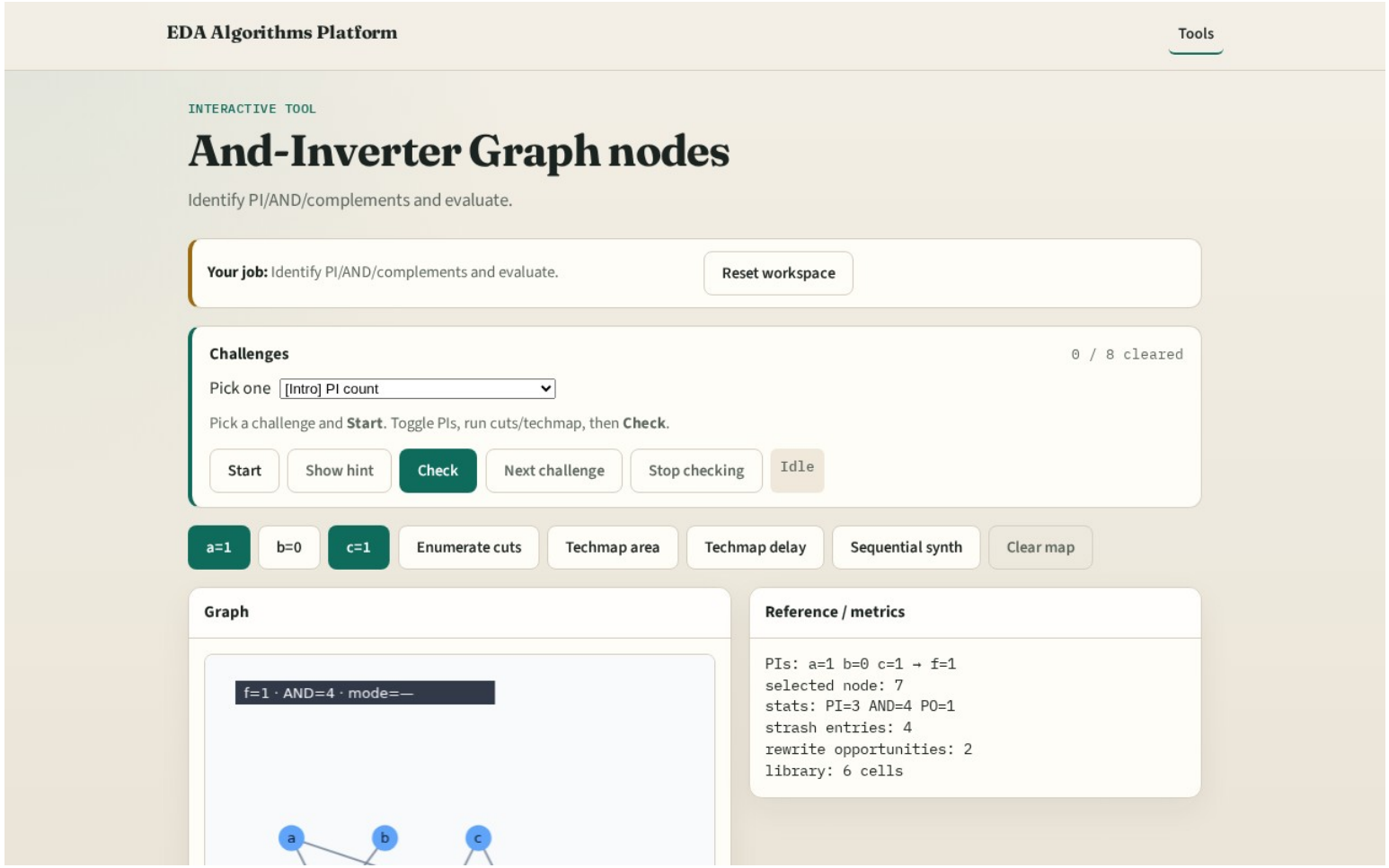
GOLDENS: 3 PI, 4 AND, 1 PO.

- Stats

```
PI=3
```

Count

Browser lab track



Browser lab starter

Implement track

- Call output_value for assignments (1,0,1) and (1,1,1)

Pitfalls

- Evaluating complemented edges wrongly
- Mixing PI names with ids

Your turn

- Next: structural hashing